

How to Better Translate Participant Quotes Using LLMs: Exploring Practices and Challenges of Non-Native English Researchers

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Abstract

Non-native English researchers face challenges in translating qualitative data, as risks of distortions and loss of nuance can affect research trustworthiness. While large language models (LLMs) provide advantages in translation, their limitations—such as inconsistent results and difficulties in capturing culturally and linguistically specific contexts—bring attention to how researchers can better integrate LLMs for accurate and contextual quote translation. To better understand this, we interviewed Korean HCI researchers about how they employ LLMs for quote translation. Our findings highlight five categorized practices used by researchers to ensure precise and comprehensible quote translation, as well as remaining challenges, including cultural bias in global LLMs and potential data leakage risks. We conclude by proposing four recommendations for researchers on using LLMs for better participants' quote translation, including effective translation strategies and ethical considerations.

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods; HCI theory, concepts and models.**

Keywords

Quote translation, Non-native English participants, Large Language Models (LLMs), Qualitative research, Scientific writing, Practices

ACM Reference Format:

Huisung Kwon, Soyeong Min, and Sangsu Lee. 2025. How to Better Translate Participant Quotes Using LLMs: Exploring Practices and Challenges of Non-Native English Researchers. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3706599.3719845>

1 Introduction

Research in global academic communities including Human-Computer Interaction (HCI) transcends borders, languages, and cultures. A

significant portion of research is conducted in countries where English is not the native language, bringing diverse perspectives and enhancing academic diversity [3, 35, 43, 51]. To share their work within the global community, researchers inevitably face the challenge of overcoming language barriers. This task goes beyond merely writing papers in English, often requiring researchers to convey the views of non-native English (NNE) participants in a way that preserves their original meanings and cultural nuances.

In academic papers, participants' quotes serve as qualitative evidence that enhances the credibility of the research [9, 50]. Translating these quotes is deeply connected to the trustworthiness, transparency, and authenticity of the research, as it is not merely a linguistic conversion but an interpretive task performed by the researcher [20, 30, 39, 49]. Previous studies have explored recommended approaches, practical guidelines for collaborating with professional translators, and researchers' attitudes toward quote translation [13, 49, 55] to ensure the **accurate and contextual translation**—conveying the intended nuances without distorting the original text.

The emergence of AI-based tools, particularly those powered by large language models (LLMs), has opened new opportunities for research. These tools also help researchers overcome language barriers, as LLMs offer high-quality translation comparable to those of human translators, while saving time and costs [14, 26]. Notably, some NNE researchers are leveraging LLMs to enhance the translation of participants' quotes. However, despite their potential, LLMs still have limitations in capturing complex contexts, often producing inconsistent results due to their generative nature, and occasionally showing lower-quality translation for underrepresented languages and cultures [7, 15, 26, 41, 52, 56]. Therefore, researchers should take a cautious approach when using LLM tools for quote translation. However, there has been limited discussion on how to use LLM tools for quote translation to achieve accurate and contextual translation.

Building on this context, this study originates from the firsthand experience of the authors, HCI researchers based in South Korea, who faced these issues when translating quotes using LLM tools. We expected that examining how researchers employ LLMs for quote translation could reveal valuable insights into their practices for achieving precise and contextually accurate translation. To explore this, we conducted individual semi-structured interviews with HCI researchers whose native language is Korean, having intermediate or higher English proficiency—capable of assessing AI-generated translations but not at a native level—and experience integrating LLMs into research. Our findings identify distinct translation practices researchers employ and highlight unsolved

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CHI EA '25, Yokohama, Japan

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ACM ISBN 979-8-4007-1395-8/25/04

<https://doi.org/10.1145/3706599.3719845>

challenges that complicate the process. Based on these findings, we discuss four recommendations for researchers on using LLMs for better participants' quote translation, focusing on effective translation strategies and ethical considerations. This research contributes to laying the groundwork for shaping future norms of using LLMs, ensuring trustworthiness and better capturing the voices of NNE participants in quote translation.

2 Related Works

2.1 Quote Translation in Academic Research

In many academic fields, NNE researchers face the task of translating qualitative data, particularly participant quotations. While quote translation has received limited attention, a few studies have consistently brought into focus the importance of quote translation and necessity for creating structured guidelines. Quote translation is not merely a linguistic conversion but an interpretive task. Since original words and English words rarely correspond on a one-to-one basis, conceptual equivalence translation is commonly employed [20, 30, 39, 49]. The process of selecting English words to reflect original nuances inherently involves researcher's interpretation, so it is deeply connected to trustworthiness of the research [1, 6, 13, 30, 49]. Therefore, scholars emphasized that ensuring transparency of the quote translation method is crucial [39, 54]. To establish structured guidelines for quote translation, prior studies pointed out specific information that should be included in the method section [13, 50], recommended collaboration practices with professional translators (e.g., avoiding family members as translators) [1, 6, 13, 50, 55], and the importance of reflexive attitudes that researchers should adopt [20, 38, 49, 55]. Some scholars have recommended the back-translation to ensure meaning equivalence—*translating translated results back to the source language and then comparing it with original text* [2]—while discouraging multi-translation loops [1, 6, 30, 49]. With the increasing use of LLMs as an alternative to machine translators and human translators, there has been little discussion on how to effectively and ethically use LLMs for quote translation.

2.2 LLMs in Qualitative Studies and Translation

Academic practices, including qualitative studies, are evolving with LLMs. Many researchers have applied LLMs across various stages, such as generating qualitative data [12, 18, 19, 24], supporting methodology design [22, 37], facilitating analysis [25, 38, 48, 53], and enhancing academic writing [8, 17, 21, 23, 47]. However, regarding inherent limitations of LLMs (e.g., black box nature and sensitivity to provided contexts) and overreliance issues, ongoing discussions focus on ethical considerations (e.g., plagiarism, data privacy), clarifying researchers' roles, and ensuring reliability of LLM-generated outputs [7, 29, 36, 41, 42]. Despite these discussions, concrete discussions on using LLMs for quote translation, closely tied to research trustworthiness and rigor, have yet to take place.

Meanwhile, LLMs have proven effective in reducing language barriers for NNE researchers [17]. LLMs excel in ensuring lexical diversity, suggesting synonyms or alternative expressions [14, 17, 21, 26], and enabling users to generate desired results through prompt engineering [26]. However, LLMs struggle with cultural nuances and context-dependent polysemous words [7, 21, 26], and

detecting subtle shifts in tone or implicit intent compared to human translators [31], particularly due to low-resource languages and cultures [15, 36, 41, 52, 56]. Previous studies highlighted low stability and inconsistency in translation results due to their probabilistic nature [7, 26]. Nevertheless, their potential in translation and academic research is undeniable. As academia works toward establishing standards for LLM use, discussions on their role in quote translation are timely.

3 Methodology

The purpose of this study was to understand how researchers use LLMs for quote translation and identify the challenges they encounter. To achieve this, we conducted individual semi-structured interviews with HCI researchers to comprehensively explore their experiences.

3.1 Participants

Our objective was to recruit HCI researchers leveraging LLMs to enhance the quality of quote translations. Participants were required to have 1) *at least three global research publications, including one as the first author*, to ensure a deep understanding of research processes and reporting qualitative evidence (e.g. participant quote) in English. They also needed to have 2) *intermediate or higher English proficiency*, who are capable of evaluating and refining LLM-generated outputs, even if they were not native-level speakers. Eligible English proficiency were identified using their official English test scores (score ranges is in Appendix A.1.2) and self-assessed CEFR levels (B1 or above). In a screening survey, information on their research experience, English proficiency, AI tool usage experience for quote translation, and demographic details were collected. After screening, a total ten researchers were selected among 23 applicants (Table 1). The selected participants ranged in age from 25 to 36 years ($M=30.5$, $SD=3.5$), with four males, six females. This study focused on researchers whose native language is Korean, acknowledging the country's distinctive cultural and technological background as part of the study's context. South Korea, with mandatory English education, ranks among the top Asian countries in English proficiency [44]. Additionally, South Korean HCI researchers are relatively accustomed to adopting the latest technologies including LLMs, and integrating them into their research practices [11].

3.2 Study Procedure

The interview consisted of three sessions, including a brief translation activity. At the beginning of the study, the moderator explained the study objectives and key observation areas, and obtained participants' consent. The first session examined participants' research experiences and overall quote translation practices. The second session explored how participants use LLMs for quote translation, including a 15-20 min activity where they were asked to translate two quotes from the interview dataset using LLMs as they normally would. This interview dataset was pre-generated by the authors through GPT and is included in the Appendix A.3. The authors observed participants' translation processes and asked follow-up questions to understand their purposes and rationales for the use strategies of LLM. Lastly, the third session addressed participants'

Table 1: Details of Participants and Their AI Tool Usage for Quote Translation.

ID	Major	Academic Status	Research Experience†	Official English Test Score	Self-assessed CEFR Level	AI Tools Primarily Used for QT	Frequency of Using AI Tools for QT††
P1	Industrial Design	Ph.D. Year 1	3 / 3 (2)	TEPS 360	B1	ChatGPT	Almost always
P2	Industrial Design	Postdoctoral	9 / 16 (5)	TOEIC 780	B2	ChatGPT, Papago*	Almost always
P3	Industrial Design	Ph.D. Year 4+	13 / 13 (6)	TOEIC 970	C1	ChatGPT	Almost always
P4	Computer Science	Ph.D. Year 4+	4 / 7 (4)	IELTS 7.5	C1	ChatGPT	Rarely
P5	Computer Science	Postdoctoral	16 / 46 (6)	TOEIC 925	C2	ChatGPT	Occasionally
P6	Culture and Technology	Ph.D. Year 3	4 / 4 (3)	TOEIC 990	C1	ChatGPT	Almost always
P7	Computer Science	M.S. Year 2	2 / 5 (2)	TOEFL 112	C2	ChatGPT	Occasionally
P8	Electrical and Electronic Engineering	Ph.D. Year 4+	4 / 7 (6)	TOEIC 920	C1	ChatGPT	Almost always
P9	Computer Science	M.S. Year 2	4 / 6 (3)	TOEFL 110	B2	ChatGPT, Papago, Claude, ClovaX**, Gemini	Occasionally
P10	Industrial Design	Ph.D. Year 3	7 / 7 (4)	TOEFL 106	C1	ChatGPT, DeepL	Almost always

† Numbers of publications including qualitative studies / Total number of publications (First-author publications)

†† The responses on the common purposes and ways participants use AI tools are detailed in the Appendix A.2.

* Papago [33] is a multilingual machine translation cloud service by Naver Corporation.

** Clova X [32] is an AI chatbot and generative AI service developed by Naver, a South Korean search company. It leveraged Korean-language content created by Naver service (community, blog, etc.) users as training data, so it is optimized for the Korean language.

(dis)satisfactions, concerns, and considerations regarding LLMs for quote translation. One author moderated the interview while another assisted with recording, documentation, and data collection. Each interview, lasting about an hour, was conducted via a video conferencing platform using semi-structured materials.

3.3 Data Analysis & Translation Procedure

The recorded interview audio and LLM prompts in the second activity were fully transcribed. The authors initially reviewed, memoed, cross-checked the data [10], and finally conducted thematic analysis following the six phases [4, 16]. Two authors collaboratively generated initial codes which were organized into four categories aligned with the research goal: 1) values participants prioritized in translation, 2) strategies for using LLMs, 3) challenges and limitations they faced, and 4) approaches to overcome them. Building on this, the authors expanded them from the interview transcripts, repeatedly reviewing and structuring the codes until themes were identified. After adding, deleting, and refining the codes, the final themes were developed.

To outline the translation procedure, transcription and analysis were conducted in Korean to preserve contextual information. Only the quotes included in the final report were translated into English. Referring to prior studies [30, 49], we aimed to ensure contextual equivalence. Two authors manually translated the quotes, and ChatGPT4o was requested to review and proofread by providing the original text and its translation under participants' consent. The translations were finalized with the agreement among all authors. Original quotes were provided in Appendix A.5 [54].

4 Findings

4.1 How Researchers Use LLMs to Translate Participant Quotes

This section explores the practices researchers in our study employed when using LLMs to translate participant quotes (Table 2). For each researcher, overall translation process (how they used LLMs and revised results) can be found in the Appendix A.4. Their

practices were categorized by five purposes for precise and contextual translations.

4.1.1 Conveying the Original Meanings with the Most Appropriate Expressions. Ensuring that the translated quotes accurately reflected the original meaning was a key priority. To achieve this, the researchers employed three main approaches. First, the researchers used LLMs to find appropriate English **phrases** that best reflected the participants' intentions. They generated multiple sentences using LLMs and combined those they believed best captured the original meaning (P1, P4, P6, P9). They also leveraged LLMs to verify alignment with the original message by comparing the English translations with the Korean text (P5, P6) or conducting back-translations (P10). When a specific phrase did not adequately capture the intended nuance, the researchers sometimes created code-mixed sentences by combining English translations with Korean explanations for unclear parts and asked LLMs to translate them (P2, P7).

"I use tools like DeepL, Google Translate, GPT and Notion... to come up with around five translation options. When I noticed common phrases or word orders, I thought, "This seems right", and combined them." (P1)

"(Input the original Korean text and its English translation together) Check if the English translation effectively captures the nuance of the original Korean. If not, suggest up to three alternatives." (P6, prompt)

Second, the researchers used LLMs to identify the best English **terms** among words with similar nuances, directly asking LLMs for synonyms rather than searching manually in Google or dictionaries (P6, P10). For domain-specific quotes, P9 trained LLMs on relevant sources to enhance terminology suggestions.

"(For uncommon topics) I provide examples from English newspapers on the same subject and (ask GPT to) use them as reference for the translation" (P9)

Third, in this process, the researchers aimed to find expressions that not only reflected the intended meaning but also sounded natural in everyday English. While they previously sought feedback from bilingual colleagues, P4, P5 and P10 now use LLMs to check

Table 2: The overview of the practices researchers employed when using LLMs to translate participant quotes.

Purpose	How Researchers Use AI Tools
4.1.1 Conveying the Original Meanings with the Most Appropriate Expressions	
To find appropriate English phrases that best reflect the original message (P1, P4, P5, P6, P7, P9, P10)	- Generating multiple sentences with LLMs and combining them. - Leveraging LLMs to compare English translation with the original Korean text or conducting back-translation. - Creating code-mixed sentences and asking LLMs to translate them.
To identify the best English terms among words with similar nuances (P6, P9, P10)	- Directly asking LLMs for synonyms. - Training LLMs on relevant sources to enhance terminology suggestions.
To find expressions that sounded natural in everyday English (P4, P5, P10)	- Asking LLMs to check whether translated expressions are common. - Using prompts such as “<i>express this in English</i>” or “<i>rephrase</i>” instead of directly asking for a “<i>translation</i>”.
4.1.2 Incorporating Dialogue Context into Quotes	
To capture implied meanings derived from the conversational flow in the quotes (P1, P2, P3, P4, P7, P8, P9, P10)	- Explaining contextual information to LLMs before translation. - Inputting complete raw data into ChatGPT for ‘context learning’. - Including preceding sentences of the quotes in the translation process and extracting only the target quotes. - Adding parenthetical explanations to clarify the meaning of demonstratives.
4.1.3 Reflecting Cultural Contexts in Quotes	
To reflect cultural context in quotation translation (P8, P9)	- Using LLMs developed in the participant’s nation. - Input extra resources reflecting the participant’s cultural context.
4.1.4 Delivering the Participants’ Voices in a Vivid Manner	
To preserve their voice, tone, and manner of speaking in the written text (P2, P4, P5, P6, P9)	- Supplying participants’ profiles and speech traits to LLMs.
To ensure liveliness (P3, P7, P9, P10)	- Directly prompting LLMs to deliver more lively translation.
To fully capture participants’ emotions (P7, P10)	- Providing detailed descriptions of the conveyed emotions to LLMs and requesting suggestions for the most appropriate words.
4.1.5 Ensuring Quotes Clearly Support the Findings	
To ensure that the translated quotes clearly support the finding arguments (P1, P3, P8, P9, P10)	- Providing contextual information from the paper. - Providing relevant paragraphs from the paper when refining quotes.

whether expressions are commonly used. P10 used prompts such as “*express this in English*” or “*rephrase*” instead of directly asking for a “*translation*”, to achieve more natural-sounding results.

4.1.2 Incorporating Dialogue Context into Quotes. Quotes are selected excerpts from the flow of conversations in the studies. To ensure that these quotes are comprehensible even when provided independently, the researchers used three practices to incorporate the conversational context into the translation. The first practice is explaining contextual information to the LLMs before initiating the translation. P3, P8, and P10 explained the study’s purpose, the topic of the conversation, and the participant’s intention in the quote to the LLMs prior to translation. Similarly, P1, P3, P9, and P10 input complete raw data into the ChatGPT. The researchers conducted this process for the purpose of ‘context learning’, so they didn’t check the responses and subsequently began translation prompting.

“The participant is talking about the moments that he felt anxious during autonomous driving.” (P3, prompt)

The second practice addressed cases where quotes contained implied meanings derived from the conversational flow. Five researchers included sentences preceding part of the quotes to translate and extracted only the target quotes (P2, P4, P7, P8, P9). P3 and P10 explained implicit meanings to the LLMs while proofreading. Third, the researchers adopted a practice to clarify the meaning of demonstratives in source text by adding explanations in parentheses (P3, P4, P10) or directly revising the original text (P7) before providing it to the LLMs.

“The phrase ‘so even just signaling’ implies that ‘even if other cars in front or nearby simply turn on their lights or blinkers.’” (P10, prompt)

4.1.3 Reflecting Cultural Contexts in Quotes. The researchers hoped that the LLMs would understand Korean culture and expressions unique to the Korean language (e.g., Did you have lunch? (How are you?), Nunchi¹). Our study identified two practices for reflecting linguistic and cultural differences during the process of translating quotes. One practice is using LLMs developed in Korea. While all researchers in this study mainly used ChatGPT, P9 also used Clova X alongside global LLMs (ChatGPT & Claude). The researcher believed that Clova X better captured Korean cultural contexts and compared translation results from these tools², as reported in Naver Inc.’s technical report [45]. When using global LLMs, the other practice was to input extra resources that encapsulate Korean cultural contexts (P8, P9). For instance, P9 searched for English articles specializing in Korean culture and used them as a training source.

“(when translating a context that is unfamiliar in Western culture) I referred to sources like the Korean Herald, which professionally translate Korean news into English, to observe their word choices. Then, I provided those words and contexts (to the GPT).” (P9)

¹Nunchi [28] (눈치, Korean word) means the subtle art of instantly gauging other people’s thoughts and feelings.

²The authors tested the Korean idiom “가는 날이 장날이라고* 그냥 돌아왔어요” (Reference translation: “**What luck!** I just came back with nothing.” This(*) is a metaphorical expression referring to an unexpected event while attempting to do something, which means *either unexpected fortune or misfortune*). In a zero-shot prompting, Clova X more accurately captured the meaning as “**It was my bad luck** that I returned empty-handed today,” than ChatGPT4o (“**I happened to visit on the busiest day** and had to return empty-handed.”)

4.1.4 Delivering the Participants’ Voices in a Vivid Manner. Since quotes are transcriptions of participants’ spoken language, the researchers emphasized the importance of vividly preserving their voice, tone, and manner of speaking in the written text. After understanding participants’ speech patterns and verbal characteristics by revisiting audio data, the researchers began supplying participants’ profiles and speech traits to LLMs to naturally reflect these traits in the translations (P2, P4, P5, P6, P9). For instance, P2 reported including information of the participant’s age group and manner of speaking during the refinement process to better capture their unique voice.

“When refining with GPT, I often check whether the nuances of the quotes align with the participant’s profile (e.g., a man in his 30s).” (P2)

Some researchers directly prompted LLMs for more lively translations (P3, P7, P9, P10). While seeking vividness, P1, P6, and P7 requested that quotes remain concise for readability. For instance, P7 asked ChatGPT to verify that the liveliness and key points were preserved while unnecessary filter words were removed to keep it concise.

“Since quotes are conversational, I pay close attention to whether it sounds colloquial in English. Sometimes, when asking (LLMs) for edits, I say, ‘This sounds too formal, make it more colloquial and casual.’” (P3)

Additionally, to fully capture participants’ emotions, the researchers reflect emotional types and intensity within quotes. P10 provided ChatGPT with detailed descriptions of the conveyed emotions and requested suggestions for the most appropriate words.

“What does ‘uneasy’ mean exactly? The participant described the emotion as ‘a state of discomfort caused by tension from uncertainty.’ Suggest 10 suitable words or phrases, and share your best pick.” (P10, prompt)

4.1.5 Ensuring Quotes Clearly Support the Arguments. Quotes serve as qualitative evidence to support the study’s argument [9, 50]. Therefore, ensuring that the translated quotes clearly support the finding arguments is an important task for researchers. Five researchers (P1, P3, P8, P9, P10) believed that providing contextual information from the paper would result in more accurate translations of quotes. For instance, P8 provided both the English content from the findings section and the original Korean quote to ChatGPT, requesting an English translation. P9 explained that the initial translation used only the surrounding conversational data, but relevant paragraphs from the paper were later added to refine the translation and align it with the argument.

4.2 Challenges Researchers Face When Using LLMs for Translating Participant Quotes

In this section, we report on two key challenges and issues that the researchers face when using LLMs for translating quotes, even while employing the practices in Section 4.1.

4.2.1 Cultural Bias in Global LLMs and Challenges for Non-Native English Researchers. Global LLMs are primarily trained by Western datasets, which limits their ability to fully reflect the cultural contexts of diverse regions [5, 15, 41, 52, 56]. The researchers noted

that while these tools understand broad Korean cultural concepts, they often fail to provide translation results that accurately capture specific contexts (P5, P8, P9, P10). P8 specifically stated that LLMs struggle to accurately translate Korean politically charged words, expressions derived from traditional folklore, and newly emerging phrases in contemporary culture (e.g. bamboo grove³). As a result, P5 avoided asking global LLMs to provide translations involving cultural nuances. To address these challenges, researchers have employed strategies such as using locally developed LLMs (refer to Section 4.1.3). However, these tools often fail to generate natural English expressions comparable to global LLMs like GPT or Claude, requiring cross-verification, which is significantly time-consuming for research tasks (P8, P9). Beyond the issue of cultural bias in global LLMs, P10 mentioned a challenge faced by NNE researchers, criticizing the English-dominated academic system. The NNE researchers bear the additional burden of performing translation tasks, often leading to inevitable misrepresentation of the original text. If LLMs better understood cultural and linguistic contexts, P10 expressed hope that they could enable seamless translation across all languages, even when studies are reported in their original language.

“I think it would be ideal if researchers could preserve the original context in their own language, and then tailor the language (of the paper) depending on who the reader is at the time.” (P10)

4.2.2 Data Leakage Risks and Challenges in Handling Sensitive Content. with LLMs As highlighted in previous research [38, 41, 42, 46, 48], the researchers have also expressed concerns about the risks of data leakage when using LLMs with study data and difficulties handling sensitive content in quotes. Four researchers (P3, P6, P7, P8) specifically expressed concerns about exposing participants’ personal information and their private, sensitive content. P3 worried that a data leak could lead to unfair plagiarism accusations against her unpublished research. P7 further wondered whether biased participant response (e.g., dislike for a specific religion) used as a training source could negatively affect data learning. P8 felt frustrated when LLMs blocked translation entirely due to sensitive content in quotes.

To mitigate data these risks, the researchers employed several strategies. Anonymization techniques were applied, such as removing personal information (P3, P7, P8, P10) and avoiding inputting extensive data with identifiable details (P3, P4). P7 used abstract expressions to mask sensitive expressions, and three researchers disabled data training on chatting logs and frequently cleared them (P3, P8, P6). However, due to a lack of trust in LLM companies, uncertainties remain about whether these logs are truly excluded from model training. The researchers also acknowledged their limited ability to address issues if a data leak occurred. Furthermore, P3 and P6 pointed out that, despite these precautions, ethical concerns remain if participants are unaware that their spoken data could be processed by LLMs for translation.

³ *Bamboo grove(forest)* [27] is referred to a channel through which a participant may disclose his/her secret to others, deriving from an old story.

5 Discussion

Our findings explored diverse researcher practices aimed at achieving accurate and contextual translation, aligned with previous studies [1, 13, 49]. With some of their practices showing potential for practical application, we discuss four recommendations for researchers on using LLMs for better participants' quote translation.

First, researchers can design prompts with detailed inputs of contextual information when requesting translations, as LLMs can produce significantly different outcomes depending on the input [7, 26]. Quotes are not standalone sentences but are closely intertwined with the conversational flow, the speaker's characteristics, and subtle nuances of expression [9]; they have a unique complexity that distinguishes them from straightforward literal translation. In our study, researchers included information about the interview setting, surrounding dialogue, cultural aspects, and participant profile details—including age, gender, and speaking style (refer to Sections 4.1.2 to 4.1.4). These strategies can be an effective approach to preserve the contextual meanings and ensure consistency across translation outputs.

Second, researchers can actively utilize LLMs to identify best expressions derived from guidelines with human translators. Previous research highlighted the need for multiple translators to reduce errors and recommended post-translation debrief sessions [1], guidelines that apply to LLM-based quote translation. Researchers can request LLMs to generate multiple translation options for a sentence and either combine or select the best one, as demonstrated in Section 4.1.1. Also, as discussed in Section 4.2.1, due to LLMs' bias toward English and Western cultures, employing a combination of local and global LLMs, where feasible, can enhance understanding of both the source language and English, reducing translation errors. Additionally, we propose that researchers can use LLMs to verify whether the translation accurately reflects the intended meaning, once the translation reaches a certain level of completion. These strategies could minimize mistranslations or loss of meaning of quote, enhancing research integrity.

Thirdly, adopting a reflexive approach in LLM-based quote translation is important. Before using LLMs, reconsidering the intervention of a human translator can be needed when: 1) the quote contains cultural contexts or implicit elements, such as tone changes, humor, or metaphors; 2) nonverbal cues are crucial to its meaning; or 3) the quote includes sensitive or provocative content so that LLM can block translation. While using LLMs, clarifying researchers' roles can be another reflexive approach. Building on Thominet et al.'s study [38], we also found out three roles in quote translation; a teacher to provide verified cultural sources to LLMs and teach contextual equivalence concept and translation quality criteria, a peer to explore nuanced word choices, and a manager to ensure consistency through techniques like back-translation [1, 26]. We recommend that researchers recognize their current role and consciously fulfill all three roles. Moreover, declaring these processes on a report is also essential, enabling researchers to further reflect on their practices and enhance the transparency of LLM-based quote translation [1, 13, 55].

Lastly, ethical awareness is crucial, particularly given the risk of data leakage and bias in LLMs. Regarding data security, we discovered several strategies such as anonymization and LLM training

opt-out settings—basic but necessary safeguards. As noted in other studies [41, 42], researchers should have the ethical responsibility to inform participants and respect their agency, like obtaining explicit consent outlining the purpose of LLM usage and measures to prevent data leakage. Additionally, researchers should remain vigilant about whether LLMs embed cultural or stereotypical biases by checking social sensitivity and adopting an iterative review process throughout the translation [40].

6 Limitations & Future Works

We acknowledge that this study was conducted with a limited sample of participants, emphasizing the need for diversity in future research. First, this study focused on a Korean sample, examining the process of translating Korean into English. As cultural context and linguistic specificity are critical to quote translation, they may limit the generalizability of our findings. Nevertheless, since our findings capture the broader landscape of translation rather than being specific to Korean, we hope that this study's case extends beyond Korean and applies to other languages. Second, the study involved researchers with intermediate to advanced English skills, who are not native-level speakers but can evaluate LLM-generated translations, making them a suitable starting point for exploring translation practices. Future research could investigate how LLMs support individuals with low English proficiency, given their potential to reduce language barriers. Lastly, as the study targeted HCI researchers familiar with AI and user studies, expanding our findings to other academic disciplines is essential, given the widespread use of qualitative research. Alongside addressing sample limitation, further validation of researchers' practices is needed to assess their effectiveness, while also reflecting on whether this process imposes an excessive burden on researchers. Future work, including validation with bilinguals and professional translators, could help develop structured guidelines for accurate, contextual quote translation.

7 Conclusion

This research was initiated by the authors' personal motivation in South Korea, who inquired about how LLMs could be used effectively and reliably when translating quotes in English. Through semi-structured interviews with ten Korean HCI researchers, we investigated how they use LLMs for quote translation to ensure semantically and contextually accurate translation and the challenges they encountered. Based on them, four recommendations for researchers on LLM-based quote translation, involving effective usage strategies and ethical considerations. This study lays the foundation for establishing norms for quote translation using LLMs, while fostering inclusivity in academia.

References

- [1] Rasmieh Al-Amer, Lucie Ramjan, Paul Glew, Maram Darwish, and Yenna Salamonson. 2015. Translation of interviews from a source language to a target language: Examining issues in cross-cultural health care research. *Journal of clinical nursing* 24, 9–10 (2015), 1151–1162.
- [2] Maria Birbili. 2000. Translating from one language to another. *Social research update* 31, 1 (2000), 1–7.
- [3] Paula Bourges-Waldegge and Stephen AR Scrivener. 1998. Meaning, the central issue in cross-cultural HCI design. *Interacting with computers* 9, 3 (1998), 287–309.

- [4] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3, 2 (2006), 77–101. doi:10.1191/1478088706qp0630a arXiv:https://www.tandfonline.com/doi/pdf/10.1191/1478088706qp0630a
- [5] HAMZA CHAUDHRY. 2024. *AI Testing Mostly Uses English Right Now. That's Risky.* https://time.com/7001812/ai-testing-english-language-risks-essay/?utm_source=chatgpt.com
- [6] Hsiao-Yu Chen and Jennifer RP Boore. 2010. Translation and back-translation in qualitative nursing research: methodological review. *Journal of clinical nursing* 19, 1-2 (2010), 234–239.
- [7] Prokopis A Christou. 2023. How to use artificial intelligence (AI) as a resource, methodological and analysis tool in qualitative research? *The Qualitative Report* 28, 7 (2023), 1968–1980.
- [8] Steven R Cooperman and Roberto A Brandão. 2024. AI assistance with scientific writing: Possibilities, pitfalls, and ethical considerations. 100350 pages.
- [9] Anne Corden and R Sainsbury. 2007. *Using verbatim quotations in reporting qualitative social research: A review of selected publications.* Social Policy Research Unit, University of York York.
- [10] John W Creswell and Cheryl N Poth. 2016. *Qualitative inquiry and research design: Choosing among five approaches.* Sage publications.
- [11] Daeil Chin, Yoon Jung, and Youbin Shin. 2024. *Korea IDC Announces Study on the Application of Generative AI in Domestic Workplaces.* https://www.idc.com/getdoc.jsp?containerId=prAP52633124&utm_medium=rss_feed&utm_source=alert&utm_campaign=rss_syndication
- [12] Andreas Dengel, Rupert Gehrlein, David Fernes, Sebastian Görlich, Jonas Maurer, Hai Hoang Pham, Gabriel Großmann, and Niklas Dietrich genannt Eisermann. 2023. Qualitative research methods for large language models: Conducting semi-structured interviews with ChatGPT and BARD on Computer Science Education. In *Informatics*, Vol. 10. MDPI, 78.
- [13] Sina K Feldermann and Martin RW Hiebl. 2020. Using quotations from non-English interviews in accounting research. *Qualitative Research in Accounting & Management* 17, 2 (2020), 229–262.
- [14] Linling Fu and Lei Liu. 2024. What are the differences? A comparative study of generative artificial intelligence translation and human translation of scientific texts. *Humanities and Social Sciences Communications* 11, 1 (2024), 1–12.
- [15] Sourjit Ghosh and Aylin Caliskan. 2023. Chatgpt perpetuates gender bias in machine translation and ignores non-gendered pronouns: Findings across bengali and five other low-resource languages. In *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society*. 901–912.
- [16] Graham R. Gibbs. 2018. *Analyzing Qualitative Data.* SAGE Publications Ltd, London. 232 pages. http://digital.casalini.it/9781526426130 Casalini id: 5017782.
- [17] Auro Del Giglio and Mateus Uerlei Pereira da Costa. 2023. The use of artificial intelligence to improve the scientific writing of non-native English speakers. *Revista da Associação Médica Brasileira* 69, 9 (2023), e20230560.
- [18] Sonakshi Gupta, Akhlak Mahmood, Pranav Shetty, Aishat Adebayo, and Rampi Ramprasad. 2024. Data extraction from polymer literature using large language models. *Communications Materials* 5, 1 (2024), 269.
- [19] Perttu Hämmäläinen, Mikke Tavast, and Anton Kunnari. 2023. Evaluating large language models in generating synthetic HCI research data: a case study. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [20] Esther Helmich, Sayra Cristancho, Laura Diachun, and Lorelei Lingard. 2017. 'How would you call this in English?' Being reflective about translations in international, cross-cultural qualitative research. *Perspectives on medical education* 6 (2017), 127–132.
- [21] Jingshan Huang and Ming Tan. 2023. The role of ChatGPT in scientific communication: writing better scientific review articles. *American journal of cancer research* 13, 4 (2023), 1148.
- [22] Bernard J Jansen, Soon-gyo Jung, and Joni Salminen. 2023. Employing large language models in survey research. *Natural Language Processing Journal* 4 (2023), 100020.
- [23] Mohamed Khalifa and Mona Albadawy. 2024. Using artificial intelligence in academic writing and research: An essential productivity tool. *Computer Methods and Programs in Biomedicine Update* (2024), 100145.
- [24] Junsol Kim and Byungkyu Lee. 2023. AI-Augmented Surveys: Leveraging Large Language Models and Surveys for Opinion Prediction. *arXiv preprint arXiv:2305.09620* (2023).
- [25] Matheus de Moraes Leça, Lucas Valença, Reyndre Santos, and Ronnie de Souza Santos. 2024. Applications and Implications of Large Language Models in Qualitative Analysis: A New Frontier for Empirical Software Engineering. *arXiv preprint arXiv:2412.06564* (2024).
- [26] Tong King Lee. 2024. Artificial intelligence and posthumanist translation: ChatGPT versus the translator. *Applied Linguistics Review* 15, 6 (2024), 2351–2372.
- [27] Korean LII. January 21, 2025. *Meaning of Bamboo Grove.* http://www.koreanlii.or.kr/w/index.php/Bamboo_grove
- [28] Korean LII. January 21, 2025. *Meaning of Nunchi.* http://www.koreanlii.or.kr/w/index.php/Nunchi
- [29] Brady D Lund, Ting Wang, Nishith Reddy Mannuru, Bing Nie, Somipam Shimray, and Ziang Wang. 2023. ChatGPT and a new academic reality: Artificial Intelligence-written research papers and the ethics of the large language models in scholarly publishing. *Journal of the Association for Information Science and Technology* 74, 5 (2023), 570–581.
- [30] Pratap Chandra Mandal. 2018. Translation in qualitative studies: Evaluation criteria and equivalence. *The Qualitative Report* 23, 10 (2018), 2529–2537.
- [31] Ahmed Mohammed Moneus and Yousef Sahari. 2024. Artificial intelligence and human translation: A contrastive study based on legal texts. *Heliyon* 10, 6 (2024).
- [32] Naver. January 21, 2025. *ClovaX.* https://clova-x.naver.com
- [33] Naver. January 21, 2025. *Papago.* https://papago.naver.com
- [34] Council of Europe. Council for Cultural Co-operation. Education Committee. Modern Languages Division. 2001. *Common European framework of reference for languages: Learning, teaching, assessment.* Cambridge University Press.
- [35] Ihudiya Finda Ogbonnaya-Ogburu, Angela DR Smith, Alexandra To, and Kentaro Toyama. 2020. Critical race theory for HCI. In *Proceedings of the 2020 CHI conference on human factors in computing systems*. 1–16.
- [36] Étienne Ollion, Rubing Shen, Ana Macanovic, and Arnault Chatelain. 2024. The dangers of using proprietary LLMs for research. *Nature Machine Intelligence* 6, 1 (2024), 4–5.
- [37] Jessica L Parker, Veronica Richard, and Kimberly Becker. 2023. Flexibility & iteration: Exploring the potential of large language models in developing and refining interview protocols. *The qualitative report* 28, 9 (2023), 2772–2790.
- [38] Mike Perkins and Jasper Roe. 2024. The use of Generative AI in qualitative analysis: Inductive thematic analysis with ChatGPT. *Journal of Applied Learning and Teaching* 7, 1 (2024).
- [39] Erika C Piazzoli. 2015. Translation in cross-language qualitative research: Pitfalls and opportunities. *Translation and Translanguaging in Multilingual contexts* 1, 1 (2015), 80–102.
- [40] Sebastin Santy, Kalika Bali, Monojit Choudhury, Sandipan Dandapat, Tanuja Ganu, Anurag Shukla, Jahanvi Shah, and Vivek Seshadri. 2021. Language Translation as a Socio-Technical System: Case-Studies of Mixed-Initiative Interactions. In *Proceedings of the 4th ACM SIGCAS Conference on Computing and Sustainable Societies (Virtual Event, Australia) (COMPASS '21)*. Association for Computing Machinery, New York, NY, USA, 156–172. doi:10.1145/3460112.3471954
- [41] Hope Schroeder, Marianne Aubin Le Quéré, Casey Randazzo, David Mimmo, and Sarita Schoenebeck. 2024. Large language models in qualitative research: Can we do the data justice? *arXiv preprint arXiv:2410.07362* (2024).
- [42] Hong Shen, Tianshi Li, Toby Jia-Jun Li, Joon Sung Park, and Diyi Yang. 2023. Shaping the emerging norms of using large language models in social computing research. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing*. 569–571.
- [43] SIGCHI. 2025. *What is SIGCHI?* https://sigchi.org/about/
- [44] Arina Sofiah. 2023. *Regions in Asia with the highest and lowest English proficiency: 2023 Index.* https://www.humanresourcesonline.net/regions-in-asia-with-the-highest-and-lowest-english-proficiency-2023-index
- [45] Guijin Son, Hanwool Lee, Sungdong Kim, Seungone Kim, Niklas Muennighoff, Taekyoon Choi, Cheonbok Park, Kang Min Yoo, and Stella Biderman. 2024. Kmmu: Measuring massive multitask language understanding in Korean. *arXiv preprint arXiv:2402.11548* (2024).
- [46] Samuel Sousa and Roman Kern. 2023. How to keep text private? A systematic review of deep learning methods for privacy-preserving natural language processing. *Artificial Intelligence Review* 56, 2 (2023), 1427–1492.
- [47] Arthur Tang, Kin-Kit Li, Kin On Kwok, Liujiao Cao, Stanley Luong, and Wilson Tam. 2024. The importance of transparency: Declaring the use of generative artificial intelligence (AI) in academic writing. *Journal of nursing scholarship* 56, 2 (2024), 314–318.
- [48] Luke Thominet, Jacqueline Amorim, Kristine Acosta, and Vanessa K Sohan. 2024. Role Play: Conversational Roles as a Framework for Reflexive Practice in AI-Assisted Qualitative Research. *Journal of Technical Writing and Communication* (2024), 00472816241260044.
- [49] Zeynep Turhan and Claudia Bernard. 2022. Challenges and strategies of translation in a qualitative and sensitive research. *Qualitative Research* 22, 3 (2022), 487–495.
- [50] Fenna Van Nes, Tineke Abma, Hans Jonsson, and Dorly Deeg. 2010. Language differences in qualitative research: is meaning lost in translation? *European journal of ageing* 7 (2010), 313–316.
- [51] Adriana S Vivacqua, Naomi Yamashita, Shaowen Bardzell, and Neha Kumar. 2021. SIGCHI Across Borders: Bringing Together the Local and Global. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing*. 238–240.
- [52] Jianhao Yan, Pingchuan Yan, Yulong Chen, Judy Li, Xianchao Zhu, and Yue Zhang. 2024. Gpt-4 vs. human translators: A comprehensive evaluation of translation quality across languages, domains, and expertise levels. *arXiv preprint arXiv:2407.03658* (2024).

- [53] Lixiang Yan, Vanessa Echeverria, Gloria Milena Fernandez-Nieto, Yueqiao Jin, Zachari Swiecki, Linxuan Zhao, Dragan Gašević, and Roberto Martinez-Maldonado. 2024. Human-AI Collaboration in Thematic Analysis using ChatGPT: A User Study and Design Recommendations. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [54] Ahtisham Younas, Sergi Fàbregues, Angela Durante, and Parveen Ali. 2022. Providing English and native language quotes in qualitative research: a call to action. *Nursing open* 9, 1 (2022), 168–174.
- [55] Nor Akma Yunus, Tim olde Hartman, Peter Lucassen, Chris Barton, Grant Russell, Areni Altun, and Elizabeth Sturgiss. 2022. Reporting of the translation process in qualitative health research: a neglected importance. *International Journal of Qualitative Methods* 21 (2022), 16094069221145282.
- [56] Xiang Zhang, Senyu Li, Bradley Hauer, Ning Shi, and Grzegorz Kondrak. 2023. Don't Trust ChatGPT when Your Question is not in English: A Study of Multilingual Abilities and Types of LLMs. *arXiv preprint arXiv:2305.16339* (2023).

A Appendix

A.1 Participant English Proficiency Criteria

A.1.1 CEFR Self-Assessment Rubrics of Writing [34]. The self-assessment grid illustrates the levels of proficiency described in the Common European Framework of Reference for Languages (CEFR) and is an essential component of the ELP.

A1. I can write a short, simple postcard, for example sending holiday greetings. I can fill in forms with personal details, for example entering my name, nationality and address on a hotel registration form.

A2. I can write short, simple notes and messages relating to matters in areas of immediate needs. I can write a very simple personal letter, for example thanking someone for something.

B1. I can write simple connected text on topics which are familiar or of personal interest. I can write personal letters describing experiences and impressions.

B2. I can write clear, detailed text on a wide range of subjects related to my interests. I can write an essay or report, passing on information or giving reasons in support of or against a particular point of view. I can write letters highlighting the personal significance of events and experiences.

C1. I can express myself in clear, well-structured text, expressing points of view at some length. I can write about complex subjects in a letter, an essay or a report, underlining what I consider to be the salient issues. I can select style appropriate to the reader in mind.

C2. I can write clear, smoothly-flowing text in an appropriate style. I can write complex letters, reports or articles which present a case with an effective logical structure which helps the recipient to notice and remember significant points. I can write summaries and reviews of professional or literary works.

A.1.2 Score Ranges of Official English Test. In the screening process for participants, only individuals who scored within or above the intermediate range were selected (Table 3).

A.2 The responses of the researchers on how they use AI tools when translating quotes

In the screening survey process, our aim was to recruit participants who use AI tools to translate participant quotes. In addition to asking how frequently they use AI tools, we included an open-ended question asking them to briefly describe the purposes and methods of their AI tool usage. The responses are detailed in the Table 4 below.

Table 3: Score Ranges of Official English Tests.

Test	Top Score	Score Range
TOIEC	990	Basic (0–450), Intermediate (451–650), Advanced (651–850), Proficient (851–990)
TOEFL	120	Basic (0–31), Intermediate (32–72), Advanced (73–94), Proficient (95–120)
TEPS	600	Basic (0–199), Intermediate (200–399), Advanced (400–499), Proficient (500–600)
IELTS	9.0	Basic (1.0–3.0), Intermediate (3.5–5.0), Advanced (5.5–6.5), Proficient (7.0–9.0)

Table 4: The purposes and methods of using AI tools in the translation process.

ID	The purposes and methods of using AI tools in the translation process
P1	I use ChatGPT to preserve the nuances of spoken language and enhance vivid expressions.
P2	First, I translate the quotes myself and use AI to check the accuracy of vocabulary and grammar. Additionally, when there are cultural nuances that differ, I ask the LLM how to best convey them.
P3	I input the entire quote into ChatGPT and request a translation to save time, and I find that it generates more natural spoken language than I do. If the initial translation doesn't fit the context, I provide additional contextual information and request a revision.
P4	I usually translate the quotes myself, but when it's difficult to convey nuances accurately in English, I ask ChatGPT for similar English expressions and choose the one that best captures the intended nuance.
P5	After translating the text into English myself, I request ChatGPT for proofreading and then further discuss Korean cultural nuances with it.
P6	I used AI for the initial translation to save time. If the translation felt too literal or unnatural, I engaged in a dialogue with ChatGPT to refine it into more natural-sounding speech.
P7	If there aren't many quotes to translate or if they are in the form of indirect quotations, I translate them myself. However, when there are many quotes and translating them becomes cumbersome, I use GPT.
P8	I instruct ChatGPT to translate quotes naturally, then review the translated quotes by comparing them with the original Korean quotes to assess their validity.
P9	I first ask the LLM to translate the text, and if the nuances are not well captured, I refine it by adding user information or referencing quotes from papers on similar topics.
P10	I used AI tools to translate into natural, conversational English.

A.3 Sample Interview Data in Translation Activity

The sample interview data was generated by asking GPT-4 to create content based on experiences with autonomous vehicles. The authors then refined the raw data by adding filler words, converting subjects into demonstrative pronouns, and shaping the text into a more realistic interview format. Subsequently, the authors selected two quotes for translation. The first quote features long-winded speech, including demonstrative pronouns, contextually ambiguous vocabulary (e.g., '켜다': turn on/operate), and phrases with little significant meaning (e.g., 'that's how it was'). The second quote is much shorter and conveys implicit meaning through the use of degree expressions (e.g., '그 정도면': for that level). Two quotes are emphasized using bold text.

Sample Interview Dataset (M: moderator, P: participant) :

M: Have you had any memorable experiences using autonomous driving mode?

P: Hmm... Ah! On the highway for about one or two hours? Yeah. I tried it then, and I definitely felt less tired. It was also nice just to rest my hands on the wheel.

M: Was there a specific reason you decided to use autonomous driving mode in that situation?

P: I do a lot of long-distance driving, and that day I had to travel far again. I was really tired that day and was debating whether to go or not. Haha... Driving while sleepy was tough, so... yeah, that's why I used it.

M: Have you ever done other activities while using autonomous driving?

P: Hmm... I wouldn't say I did many things. It's more like I looked out the window and listened to music? I'd never had a chance to look around while on the highway, but being able to do that was nice. It made me think, "I can rest even while driving."

M: Were there any moments where you felt uneasy while using autonomous driving?

P: Oh, yes, there were. I was just cruising along with no cars ahead, feeling pretty relaxed. A car behind me was speeding, so I turned on the signal to let it pass, but suddenly the autonomous system tried to slow down, which startled me. **It doesn't understand that kind of implicit communication between people, so (I) was anxious even when (I) tried to turn/operate on... that's how it was.** (Original: 이게 사람끼리 하는 그 암묵적인 의사소통을 애는 모르니까 그래서 뭐 켜려고만 해도 막 불안해지고.. 그랬습니다.)

M: Did you continue using autonomous driving after that?

P: I turned it off for a bit. I started to lose trust in it and felt tense, so I thought it'd be better if I just drove myself.

M: Have you had other experiences where you switched to manual mode while using autonomous driving?

P: Hmm... yeah. One time, I tried it while driving in the city. It didn't detect something suddenly popping out in an alley, so I hit the brakes myself. After that, I just stopped expecting much and switched to manual driving.

M: Did the system give you a warning sound or respond in that situation?

P: Oh, a warning sound? **(It) did go off, but by then(literal translation: for that level) it was too late.** (Original: 아 경고음 이요? 뜨긴 땀쥌. 근데 그정도면 너무 늦잖아요) What's needed is an immediate response... When I drive, I have basic data in my head, so I can anticipate things and adjust my speed or scan my surroundings. But AI doesn't seem capable of that. That's actually the most important part of driving... so it made me think there's still a long way to go.

M: After that, have you avoided using autonomous driving in urban areas?

P: Yeah, I think it's still much better for me to drive in the city. It's still a long way off. Highways don't have as many varying situations, but in the city, things change too quickly.

A.4 Participants (Researchers)' Quote Translation Process

The overall translation process of participants (researchers), including how they used LLMs and revised the results, is detailed in Figure 1.

A.5 Participant Quotes in Original (Korean) Language in Section 4

As we emphasize preserving the original meaning when translating participants' quotes as qualitative evidence, we provide the original Korean text of the participants' quotes in this paper in Table 5.

Researchers' Quote Translation Process with The Number of LLMs They Used and Rounds of Revisions

ID	LLM models	Revision	Translation Process & Results
P1	1 (GPT)	1	Provide research context* Provide study data Request translation Change to colloquial word Open new chat & regenerate answers Back-translation Translated Quote It doesn't get that unspoken understanding people have with each other, so I'd get nervous even just trying to signal something. It made me uneasy.
P2	1 (GPT)	3	Manual translation Provide participant profile Change to nuanced word Reflect emotion/status Refine details manually Translated Quote It (The system) doesn't really pick up on implicit signs that humans do. So... I was, um, kind of nervous even when I was just about to do something.
P3	1 (GPT)	2	Provide research context Request translation Regenerate answers Change Phrase Refine details manually Translated Quote I think it's because it doesn't understand the kind of unspoken communication people have with each other. So even just trying to signal something made me feel uneasy...
P4	1 (GPT)	3	Manual translation (M) Proofread (Ludwig) Request translation & compare with M Change to nuanced word Change Phrase Refine details manually Translated Quote It (the self-driving system) does not understand the implicit communication between people, so even trying to turn something on made me feel anxious...That's how it was.
P5	1 (GPT)	1	Manual translation Ask proofread (GPT) Ask reason for proofread content Change Phrase Translated Quote You see, she [Sysname] doesn't understand the implicit communication between people. So, I would get nervous even when trying to do simple things.
P6	1 (GPT)	5	Machine translation Refine details manually Web search for word usage Ask for word suggestions Paraphrase translated results → Web search for word usage Change word Add filler words to reflect emotion Remove filler words Translated Quote It doesn't understand the subtle communication between people, so just trying to turn it on made me nervous.
P7	1 (GPT)	4	Provide research context Request translation Change Phrase Change Phrase Revise the original sentences → Request Translation with code-mixed sentences (revised Korean part and other English translated parts) Add adverb to reflect emotion Translated Quote Since autonomous vehicles don't understand the implicit communication that happens between people, I felt quite anxious even when I was about to try something small.
P8	1 (GPT)	1	Provide research context Provide participant's emotion and context Add sentences preceding quotes Request translation Refine details manually Translated Quote It doesn't understand that kind of implicit communication between humans, so even just thinking about trying to manipulate something made me feel anxious.
P9	3 (GPT, Claude, ClovaX)	2	Provide study data Provide research context Request translation to GPT(G) Same Process to Claude (C) and Clova X (X) → Add sentences preceding quotes Request translation to G Same Process to C & X Manually combine the results by comparing those from G, C, and X → Remove adverb Ask G to proofread the result in a natural tone Same Process to C & X Change to colloquial word Translated Quote Since autonomous vehicles don't understand the implicit communication that happens between people, I felt quite anxious even when I was about to try something small.
P10	1 (GPTs)	2	Provide research context Provide study data Translate whole study data Provide participant's emotion and context → Request translation (only quote) in a natural tone Compare two results Change to colloquial word Ask to reflect situation context → Provide implicit meaning of quote Ask nuance of emotion word Ask for word suggestions Back-translation Ask to check and proofread finally → Add adverb to reflect emotion Translated Quote It doesn't pick up on that unspoken understanding between drivers, so even when other cars turned on their blinkers or brake lights, it really put me on edge.

Meanings of color blocks

Color	Explanation
	Red color-blocks mean that the researchers provide information and context related to the target quote to LLMs. (ex. provide study data, provide participant's emotion and context) * Provide research context: information regarding the research background—such as 'This is HCI research', 'This is an interview data', and 'the paper is about [topic]' —is provided.
	Green color-blocks are commands that the researchers ask to LLMs. (ex. request translation, ask for word suggestions)
	Blue color-blocks are revisions that the researchers conduct. (ex. change to nuanced word, add adverb to reflect emotion, change phrase)
	Yellow color-blocks are validation process to check the translated results. (ex. back-translation, ask proofread)
	Gray color-blocks are extra process that the researchers conduct. (ex. manual translation, compare two results, web search for word usage)

Figure 1: Researchers' Quote Translation Process with LLMs

Table 5: Participant Quotation in This Study with Original Language and English.

Section	Language	Quote
4.1.1	Original	“DeepL, 구글 번역기, GPT, 노션으로도 번역하고... 여러가지 후보를 한 5개 짚아두고 보면서, 많이 나온 어순이나 단어면 ‘이게 맞겠지’라고 생각하고 잘 조합해서 문장을 완성시켰던 것 같아요.” (P1)
	English	“I use tools like DeepL, Google Translate, GPT and Notion... to come up with around five translation options. When I noticed common phrases or word orders, I thought, “This seems right”, and combined them.” (P1)
4.1.1	Original	“이 인용구에 대해서도 영어 번역이 한국어 원문의 뉘앙스를 잘 살리고 있는지 판단하고, 다른 대안이 있으면 3개정도 제시해줘” (P6, prompt)
	English	“(Input the original Korean text and its English translation together) Check if the English translation effectively captures the nuance of the original Korean. If not, suggest up to three alternatives.” (P6, prompt)
4.1.1	Original	“같은 주제의 영문 신문 같은 것들에서 예시를 (AI에게) 주면서 참고해서 번역해라고 요청해요.” (P9)
	English	“(For uncommon topics) I provide examples from English newspapers on the same subject and (ask AI to) use them as reference for the translation” (P9)
4.1.2	Original	““so even just signaling” 여기서 의미가 “앞 차나 주변 다른 차들이 불이나 깜빡이 같은 걸 켜기만 해도” 라는 의미를 내포하고 있어” (P10, prompt)
	English	“The phrase ‘so even just signaling’ implies that ‘even if other cars in front or nearby simply turn on their lights or blinkers.’” (P10, prompt)
4.1.3	Original	“부모님들이 그 자녀들의 결혼을 돕는 그런 맥락이 필요했는데 [...] 코리아 헤럴드 이런 것처럼 전문적으로 한국 뉴스를 영어로 번역하는 데서 어떤 식으로 어떤 단어들을 선택했는지 보고 그 단어들이랑 맥락을 (AI에게) 좀 줬던 것 같아요” (P9)
	English	“(when translating a context that is unfamiliar in Western culture) I referred to sources like the Korean Herald, which professionally translate Korean news into English, to observe their word choices. Then, I provided those words and contexts (to the AI).” (P9)
4.1.4	Original	“GPT 에 교정을 넣을 때, 인용구의 뉘앙스와 말하는 사람의 프로파일 좀 일치하는지를 좀 많이 물어보는 것 같아요 (P2)
	English	“When refining with GPT, I often check whether the nuances of the quotes align with the participant’s profile (e.g., a man in his 30s).” (P2)
4.1.4	Original	“인용구는 구어체잖아요. 그래서 영어로도 구어적인 표현인지 아닌지를 제가 좀 신경을 많이 쓰는 편인데, 가끔 (인용구를) 교정하거나 수정을 요청할 때도 ‘이거 너무 문어체니까 더 colloquial 하고 casual 하게 들리게 다시 써줘’라고 요청해요.” (P3)
	English	“Since quotes are conversational, I close pay attention to whether it sound colloquial in English. Sometimes, when asking (LLMs) for edits, I say, “This sounds too formal, make it more colloquial and casual.” (P3)
4.1.4	Original	““uneasy’가 정확하게 어떤 감정이야? 여기서 참가자가 말한 감정은 ‘무슨 일이 일어날지 몰라 긴장되어서 불편한 상태’야. 이 상황에 적절한 단어나 phrase를 10개정도 추천해줘. 그 중에 너의 best pick을 알려줘.” (P10, prompt)
	English	“What does ‘uneasy’ mean exactly? The participant described the emotion as ‘a state of discomfort caused by tension from uncertainty.’ Suggest 10 suitable words or phrases, and share your best pick.” (P10, prompt)
4.2.1	Original	“이 오리지널한 콘텍스트를 살릴 수 있는 문장들이 연구자가 생성을 하고 그 연구자의 언어로, 그다음에 이것을 읽는 사람이 누군지에 따라서 그때그때 (논문의 언어가) 바뀌면 제일 좋겠다라는 생각이 좀 드네요. (P10)”
	English	“I think it would be ideal if researchers could preserve the original context in their own language, and then tailor the language (of the paper) depending on who the reader is at the time” (P10)